SEMICONDUCTOR / PLASMA DIAGNOSTICS · 2027 GRADUATE

Plasma diagnostics and model evaluation.

From SNU optical-emission analysis to transparent stress testing of synthetic monitoring models. Focused on method, failure modes and responsible conclusions.

312
Ar viewing channels analysed via established CT-OES framework

2,808
Nine-line spectral fitting tasks (118 QC review flags)

412 pixels
Retained under stated model-screening criteria; not independent validation

PRIMARY CASE · SNU CT-OES MEASUREMENT AND RECONSTRUCTION

PROBLEM

Interpret line-of-sight Ar optical-emission measurements as spatial information without mistaking a good spectral fit or small projection residual for validated physical conditions.

MY CONTRIBUTION

Applied the laboratory's pre-existing Python CT-OES framework; organised inputs and exclusions, reviewed Gaussian fitting/QC, examined 31 × 31 reconstruction results and evaluated modelling assumptions.

RESULTS

Reviewed 312 views and nine lines (2,808 fits), with 118 QC flags. Retained 412 pixels for model-conditional Te reporting on a 31 x 31 grid. The noiseless phantom had 29.23% field error.

LIMITATIONS & DECISION

A spatial map is not a direct measurement. The 2.176 eV median depends on atomic data and radiative assumptions; no independently verified absolute density or production-line performance is inferred.

Engineering value: I can analyse messy experimental data while keeping numerical output, instrument limitations and inferred physical quantities distinct.

MICRON KOREA / TAIWAN | MEMORY PROCESS ENGINEERING | PROCESS DIAGNOSTICS

SNU raw research inputs and original CT code are not publicly redistributed. Methods and selected aggregate results appear in the accompanying technical portfolio.

SUPPORTING CASE · PLASMASENSE EVALUATION

02

Monitoring models and alarm performance.

Distinct synthetic Ar stress experiments and public measured N_2 threshold-calibration comparisons.

PROBLEM

Methods performing well under independent simulated noise can fail under shared-channel disturbances; good ranking AUC does not ensure stable thresholded alarms.

MY CONTRIBUTION

Developed reduced-physics synthetic comparisons with parent-grouped PCA/SVM splits, fixed-model covariance stress tests, and a separate public N_2 retrospective label-assisted recalibration and guard analysis.

RESULTS

Synthetic noise-aware PCA: mean FPR 1.00% under IID σ=.01 versus 82.33% under global-shared ρ=.9. N_2 label-assisted recalibration: mean FPR 50.00% → 13.56%, recall 61.09% → 99.39% on matched future frames.

LIMITATIONS & DECISION

Synthetic states are not fab failures. A 3% support guard reduced measured-case mean FPR to 9.97% but withheld 46.83% of leak frames, reducing unconditional recall to 52.56%. Retain a review flag, not an automatic veto.

SKILLS DEMONSTRATED

PYTHON · OES / SPECTRAL FITTING · CT RECONSTRUCTION · MODEL VALIDATION · PCA / SVM · CALIBRATION · DATA PROVENANCE

Scope & attribution. Existing SNU CT infrastructure and my diagnostic contribution are distinct. Ar labels are synthetic; the measured N2 reference uses dataset normal labels, not blind transfer. Three physical sessions were reused in six roles. Broader AUC 0.998915 and matched-calibration AUC 0.998360 refer to different scopes.

TECHNICAL PORTFOLIO

Full technical portfolio: available from the application website.

Public N2 data: doi.org/10.17632/9gy9v3h4mp.1 (CC BY 4.0)